1D CNN — Architecture Variant Comparison

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Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Total trials	325 (train pool 292, hold-out 33)
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	36 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	per_trial_zscore
CV folds · shuffle	7 · True
Augmentation	on · copies=6 · time_shift, time_warp, time_mask, amplitude_scale, baseline_offset, channel_dropout
Epochs · batch size	60 · 16

Sensor grid (✓ enabled, ✗ disabled, — not present)

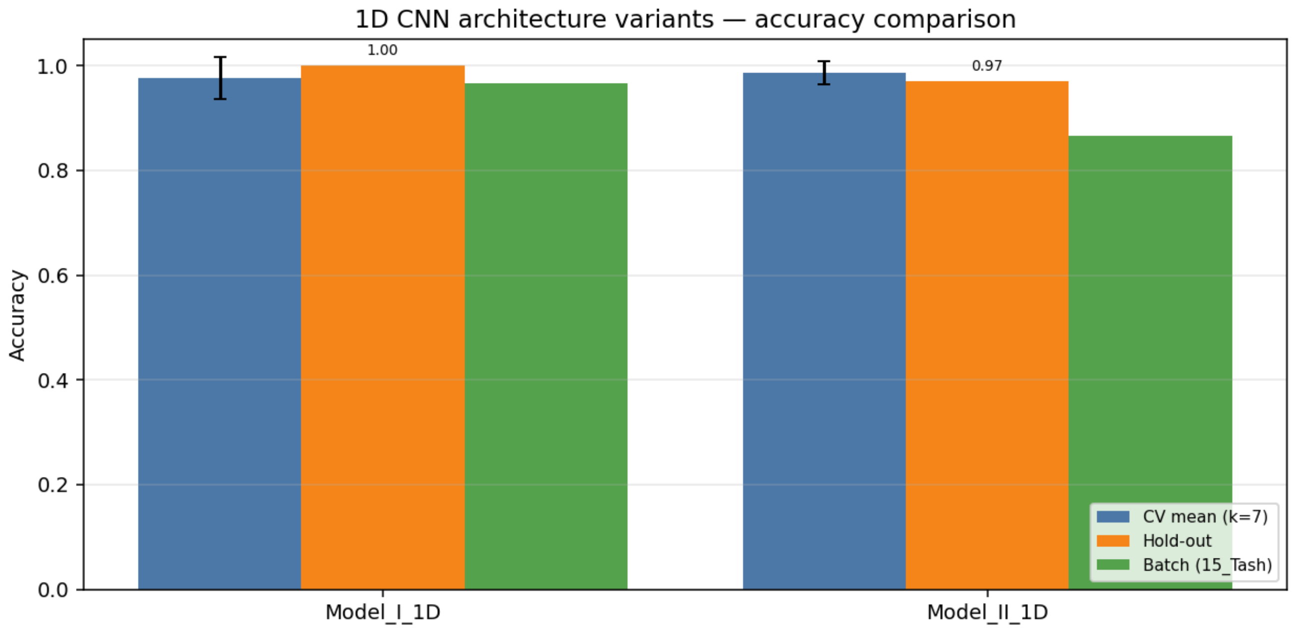
Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✗	✓	—	—
left thumb	✗	✓	✗	✓
left index	✗	✓	✗	✓
left middle	✗	✓	✗	✓
left ring	✗	✓	✗	✓
left pinky	✗	✓	✗	✓
left wrist	✗	—	—	—
right palm	✗	✓	—	—
right thumb	✗	✓	✗	✓
right index	✗	✓	✗	✓
right middle	✗	✓	✗	✓
right ring	✗	✓	✗	✓
right pinky	✗	✓	✗	✓
right wrist	✗	—	—	—

Enabled: 22 · Disabled: 24 · Modalities: YPR → 36 channels total.

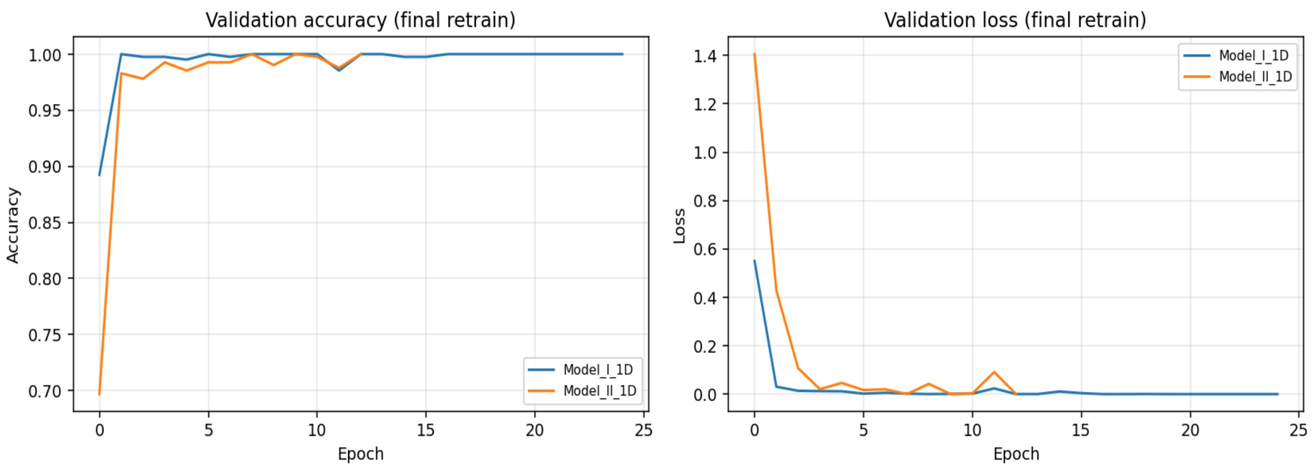
Variant comparison (summary)

Variant	Params	CV mean (k=7)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Model_I_1D	121,190	0.9762	0.0402	1.0000	0.0000	0.9667	29/30
Model_II_1D	76,486	0.9864	0.0215	0.9697	0.0372	0.8667	26/30

Best on hold-out: **Model_I_1D** · Best on CV: **Model_II_1D**



Training curves (final retrain on full training pool)



Variant: Model_I_1D

1-D adaptation of published Model I:
Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax

Trainable params	121,190
CV mean ± std	0.9762 ± 0.0402
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.9667 (29/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1750	42	1.0000	1.0000	0.9762
2	1750	42	1.0000	1.0000	1.0000
3	1750	42	0.9762	0.9762	0.9524
4	1750	42	0.9762	0.9762	0.9762
5	1750	42	0.8810	0.9524	0.9286
6	1757	41	1.0000	1.0000	0.9756
7	1757	41	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	4	5	0.800
Drum_Roll	5	5	1.000
Overall	**29**	**30**	**0.967**

Variant: Model_II_1D

1-D adaptation of published Model II (CNN+LSTM):
Conv1D(32,k=5,s=3)→Conv1D(32,k=3)→BN→Pool→Drop→LSTM(64,seq)→Drop→LSTM(64)→Drop→Dense(128)→BN→Drop→Softmax

Trainable params	76,486
CV mean ± std	0.9864 ± 0.0215
Hold-out test acc / loss	0.9697 · 0.0372
Batch-test acc	0.8667 (26/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1750	42	1.0000	1.0000	0.9762
2	1750	42	1.0000	1.0000	1.0000
3	1750	42	0.9524	0.9762	0.9524
4	1750	42	1.0000	1.0000	1.0000
5	1750	42	0.9524	0.9762	0.9524
6	1757	41	1.0000	1.0000	0.9756
7	1757	41	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	0.833	0.909	6
Double_Raise	0.857	1.000	0.923	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*0.976*	*0.972*	*0.972*	*33*
weighted avg	*0.974*	*0.970*	*0.969*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	4	5	0.800
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	5	5	1.000
Overall	**26**	**30**	**0.867**